Visualizing eDisk Models using Residual Plots

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INTRODUCTION

Protoplanetary disks are rotating "disks" of matter and gas surrounding a protostar(a young stellar object). The matter forming these disks hold information regarding what will be formed and how many. Our work involves visualizing the content of these disks using radio data collected from the **Atacama Large Millimeter/submillimeter Array (ALMA)** to detect hints to how these protostars formed. The visualization process involves running different models on each stellar object for different perspectives on the source. We use a residuals plot, the model data subtracted from the raw data, in order to depict hidden structures in the protoplanetary disk, like a **ring**, **gap**, or **spiral**. These hidden substructures reveal more about the formation process of the stellar object and allow us to infer different details about it as well.

MODELS

- **What models do we use?**

We usually start with a **Gaussian** model, which only has a few parameters and runs fast. This can help us quickly capture those basic parameters, like the position, inclination, and position angle.

Then, we go to an **exponentially tapered power law** model, which has a higher central concentration and steeper drop at larger radii. This usually fits the protoplanetary disk much better than Gaussian. A gamma-2 factor can be introduced for better precision with the power law, and fits the data better.

Based on the residuals, we can introduce more structures into our model, like a Gaussian gap to fit any potential gap structure in the disk, or an centered Gaussian. All of these would based on an **exponentially tapered power law**.

- **Which source will be primarily discussed?**

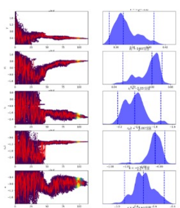
We chose to highlight sources TMC1-A and OphIRS63 since their models produced very distinctive substructures.

METHOD

- **How does our code work?**

The core of our code is a **dynamic nested sampling module** named Dynesty. The sampler would keep generating data points for each parameter of the model based on the priors we set, and comparing the model with observed data, calculating their **likelihood**, and its goal is to find the parameters that provides the greatest likelihood between the model and the data, which are, the best fit parameters. The data here is Visibility, and all these processes are done in the **UV-space** (Fourier transformed space).

Then, using data – model = residuals, and inverse transforming back to the XY-space, we can get a nice residuals image that is expected to reveal any remaining substructures that can be the evidence of planet formation.

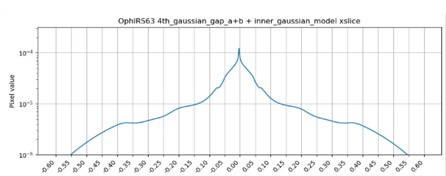


The code generates a trace plot for each parameter and keeps updating it with every additional sample. If the data did not converge to a point, we could rerun it with a different set of priors. Source: TMC1-A

- **How do we analyze the result?**

We visualize the residuals image in CASA, and plot the **contour map** based on our measurement of the background intensity. This allows us to make the structure in residuals more apparent.

We also print out the native model, slice and plot it along its major axis. This allows us to analyze and understand the profile of our model.



RESULTS

- **TMC-1A**

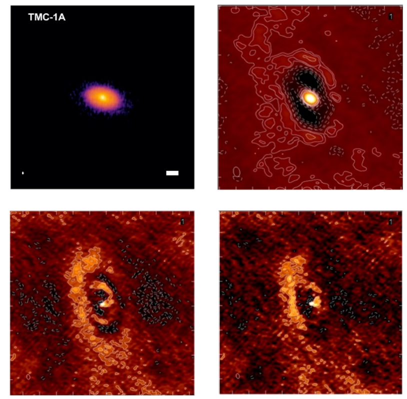


Figure 1 is the observed image of TMC1A. Figure 2-4 are residuals of model Gaussian, exponentially tapered power law (2-gamma), and power law + gap + central Gaussian. It can be seen that a ***spiral structure*** *occurs near the center. *Note that the residuals is flipped and rotated compared with the residuals.*

- **Oph IRS63**

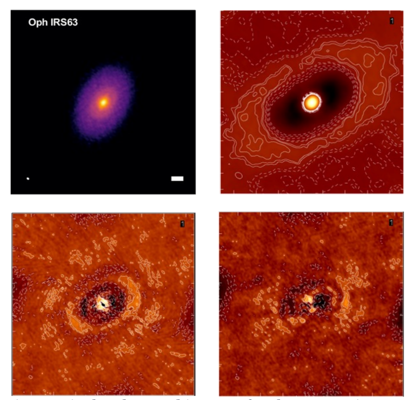


*Figure 1 is the observed image of Oph IRS63. Figure 2-4 are residuals of model Gaussian, exponentially tapered power law (2-gamma), and power law + gap + central Gaussian. Oph IRS63 is a source with two known rings. *Note that the residuals is flipped and rotated compared with the residuals.*

FUTURE WORK

We plan on completing the models for the rest of the sources in order to begin summarizing our findings in a paper. This includes reworking the code for the models we do have to fit binaries, as there are two central sources we need to account for. In order to keep expanding and gathering more data and perspectives, we also create new models based on **an exponentially tapered power law** and add more sources to diversify our collection of data models and see which sources fit our models the best.

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